In this final section, we intend to state a central limit theorem using a notion of distance that is stronger than weak convergence.

In the classical central limit theorem, we have the following.

If $S_n = X_1 + \cdots + X_n$ is the sum of n independent random variables with the same distribution (i.i.d.), and $\mathbb{E}X^2 < \infty$, then

$$\text{Law}\left(\frac{S_n - \mathbb{E}S_n}{\sqrt{\text{Var}(S_n)}}\right) \xrightarrow{w} \mathcal{N}(0,1) \qquad \text{as } n \to \infty.$$

This means that the shape of the distribution tends to the normal distribution: convergence in distribution.

There is a point to note here. The rate of convergence is not specified; one distribution may become close to the normal distribution much faster than another. This convergence does not provide us with more information, since there is no metric or numerical distance in this kind of convergence that would allow such comparisons. Closeness is a qualitative notion!

Tanaka's central limit theorem

If $S_n = X_1 + \cdots + X_n$ is a sum of independent, identically distributed random variables (i.i.d.), and $\mathbb{E}X^2 < \infty$, then

$$\text{Law}\left(\frac{S_n - \mathbb{E}S_n}{\sqrt{\text{Var}(S_n)}}\right) \xrightarrow{W_2} \mathcal{N}(0,1) \qquad \text{as } n \to \infty.$$

Thus, as n increases, the Wasserstein distance of order two from the distribution above to the standard normal distribution tends to zero. Observe that here distance is quantitative and gives a value, as a distance, for any two distributions.

To prove this theorem, we follow these steps.

Using an optimal coupling, we calculate the Wasserstein distance from an arbitrary distribution, such as X, to the normal distribution. We then use a theorem: "For two independent random variables with nonnormal distributions, the Wasserstein distance of order two of their sum is strictly less than the sum of their individual distances." Using this, we show recursively that the distance from the normal distribution for sums of the variables becomes smaller and smaller as their number increases, and tends to zero.

For simplicity in the rest of the proof, instead of using the square root of the squared distance, W_2, we work directly with the squared distance, namely $\mathbb{E}|X-Y|^2$.

Reminder. In the preceding section, we saw, without proof, that the Wasserstein distance of order two between random variables X, Y with distributions μ_1, μ_2 can be calculated using quantile functions: the optimal coupling. The square of this distance is

$$W_2^2(X,Y) = \mathbb{E}(Q_1(U) - Q_2(U))^2. \tag{1}$$

Here $U \sim \text{Uniform}(0,1)$, and Q_1, Q_2 are the quantile functions of μ_1, μ_2, respectively.

For convenience, we use the notation $W_2^2(X,Y)$ for the Wasserstein distance of order two between the distributions of the random variables X, Y, that is, μ_1, μ_2.

Theorem

Let X, Y be independent random variables with distributions f_1, f_2 and variances σ_X^2, σ_Y^2, such that $\mathbb{E}X = \mathbb{E}Y = 0$. Also define

$$\begin{aligned} Z_X &:= \mathcal{N}(0, a\sigma_X^2), \\ Z_Y &:= \mathcal{N}(0, b\sigma_Y^2), \\ Z &:= \mathcal{N}(0, a\sigma_X^2 + b\sigma_Y^2) = aZ_X + bZ_Y. \end{aligned}$$

For every $a, b \neq 0$, we have

$$W_2^2(aX+bY, Z) < a^2 W_2^2(X, Z_X) + b^2 W_2^2(Y, Z_Y)$$

Annotation above the inequality: "Strictly less."

Unless X, Y are both normal.

Meaning. The Wasserstein distance of order two from a linear combination of two independent random variables to the normal distribution with the corresponding variance is less than or equal to the sum of the distances of each of them from a normal distribution with its own variance. Equality holds only when X, Y are both normal. In other words, a linear combination of distributions becomes closer to the normal distribution.

By definition, W_2 is the infimum of the cost function over all couplings. Hence for any arbitrary coupling of X, Y, we have $W_2^2(X,Y) \leq \mathbb{E}|X-Y|^2$. Consequently,

$$\begin{aligned} W_2^2(aX+bY, Z) &\leq \mathbb{E}|aX+bY-Z|^2 \\ &= \mathbb{E}|a(X-Z_X) + b(Y-Z_Y)|^2. \end{aligned}$$

$$X - Z_X = h(X), \qquad Y - Z_Y = g(Y),$$

$$X \perp Y \implies \mathbb{E}\big[(X-Z_X)(Y-Z_Y)\big] = 0.$$

Thus,

$$W_2^2(aX+bY) \leq a^2\mathbb{E}|X-Z_X|^2 + b^2\mathbb{E}|Y-Z_Y|^2.$$

$$\longrightarrow \quad W_2^2(aX+bY,Z) \le a^2 W_2^2(X,Z_X) + b^2 W_2^2(Y,Z_Y).$$

It now remains to show the equality case. Suppose that equality holds:

$$W_2^2(aX+bY,Z) = a^2 W_2^2(X,Z_X) + b^2 W_2^2(Y,Z_Y).$$

Using (1),

$$\begin{aligned}(*)\quad & \mathbb{E}\big(Q_{aX+bY}(U) - Q_Z(U)\big)^2 \\ & = a^2 \mathbb{E}\underbrace{\big(Q_X(U_1) - Q_{Z_X}(U_1)\big)^2}_{h(U_1)} + b^2 \mathbb{E}\underbrace{\big(Q_Y(U_2) - Q_{Z_Y}(U_2)\big)^2}_{g(U_2)}.\end{aligned}$$

$$\begin{aligned}(**)\quad & \mathbb{E}\Big(\big(aQ_X(U_1)+bQ_Y(U_2)\big) - \big(aQ_{Z_X}(U_1)+bQ_{Z_Y}(U_2)\big)\Big)^2 \\ & = a^2\mathbb{E}\big(h(U_1)\big) + b^2\mathbb{E}\big(g(U_2)\big) + 2ab\mathbb{E}\big(h(U_1)g(U_2)\big).\end{aligned}$$

$$h(U_1) \perp h(U_2) \quad \Longrightarrow \quad 2ab\mathbb{E}\big(h(U_1)g(U_2)\big) = 2ab\mathbb{E}\big(h(U_1)\big)\mathbb{E}\big(g(U_2)\big).$$

It is also easy to check, using the properties of CDFs and the quantile function, that

$$(***) \qquad \mathrm{Law}\big(Q_X(U)\big) = \mathrm{Law}(X).$$

Hence,

$$\mathbb{E}\big(h(U_1)\big) = \mathbb{E}\big(g(U_2)\big) = 0.$$

Thus we can set the right-hand sides of (*) and (**) equal.

By property (***), we can now write

$$aQ_{Z_X}(U_1) + bQ_{Z_Y}(U_2) \overset{\text{distribution}}{=} aZ_X + bZ_Y = Z = Q_Z(U).$$

Substituting the right-hand side of (**) into (*) and using the preceding changes, we obtain the optimal coupling:

$$\begin{aligned}&\mathbb{E}\big(Q_{aX+bY}(U) - Q_Z(U)\big)^2 \\ &\qquad = \mathbb{E}\big(aQ_X(U_1) + bQ_Y(U_2) - Q_Z(U)\big)^2.\end{aligned}$$

$$aQ_X(U_1) + bQ_Y(U_2) = Q_{aX+bY}(U).$$

Note that U is not independent of U_1, U_2. Otherwise, the relation above would become a distributional relation.

It can be shown that the relation above holds only when X, Y both have normal distributions.

We do not prove this point here. However, we can understand it by showing that the relation above is a linear function, and that only a normal random variable has a linear quantile function.

Now consider $\{X_n\}$ to be a sequence of i.i.d. random variables with mean 0 and variance 1. Define

$$\gamma_n = \frac{X_1 + \cdots + X_n}{\sqrt{n}}.$$

For $Z \sim \mathcal{N}(0,1)$, we show that

$$W_2^2(\gamma_n, Z) \to 0 \qquad \text{as } n \to 0.$$

In this proof, we use the additional assumption $\mathbb{E}X_i^4 < \infty$. This is not one of the theorem's assumptions; it is stated here to simplify the proof.

$$\begin{aligned}
\mathbb{E}\gamma_n^4 &= \frac{1}{n^2}\mathbb{E}\left(\left(\sum_{i=1}^n X_i\right)^4\right) \\
&= \frac{1}{n^2}\sum_{i=1}^n \mathbb{E}X_i^4 + \frac{3}{n^2}\sum_{i\neq j}\mathbb{E}[X_i^2]\mathbb{E}[X_j^2] \\
&= \frac{1}{n}\mathbb{E}X_i^4 + \frac{3}{n^2}n(n-1)\underbrace{\big(\mathrm{Var}(X_i)\big)^2}_{1}.
\end{aligned}$$

Therefore,

$$\mathbb{E}\gamma_n^4 = \frac{1}{n}\mathbb{E}X_i^4 + 3\left(1 - \frac{1}{n}\right) < \text{const},$$

by the additional assumption.

To simplify the proof, in view of Theorem 2, we first examine the case $n = 2^k$. Define

$$\eta_k := \gamma_{2^k}.$$

By the second theorem, it is clear that, as $k \to \infty$, $W_2^2(\eta_k, Z)$ converges monotonically to a number, say ℓ.

Denote the limiting distribution of η_k as $k \to \infty$ by μ, and consider two independent random variables η, ξ with distribution μ. We have

$$W_2^2(\eta, Z) = \lim_{k\to\infty} W_2^2(\eta_k, Z) = \ell.$$

It is because the fourth moment is finite that we can take the limit inside W_2^2.

Also, since η is the limit of the sequence $\{\eta_k\}$ at infinity, the distance converges to ℓ:

$$W_2^2\left(\frac{\eta + \xi}{\sqrt{2}}, Z\right) = \ell.$$

Thus,

$$W_2^2\left(\frac{\eta + \xi}{\sqrt{2}}, Z\right) = W_2^2(\eta, Z).$$

Therefore η must be normal, which means that

$$\ell = 0.$$

It now suffices to extend the preceding result to all n, not only $n = 2^k$. Write $\widetilde{n} \geq 1$ in the form

$$\widetilde{n} = \sum_{k=0}^{m} \varepsilon_k 2^k, \qquad \varepsilon_k = 0 \text{ or } 1.$$

This is the binary expansion of a number. Thus every n can be written as blocks whose sizes are of the form 2^k. By adding these blocks, Theorem 2 gives

$$W_2^2(\gamma_n, Z) \leq \frac{1}{\widetilde{n}} \sum_{k=0}^{m} \varepsilon_k 2^k W_2^2(\eta_k, Z).$$

Now, as k increases, we see that, for terms with large k, $W_2^2(\eta_k, Z)$ tends to zero. Also, $\widetilde{n}$ increases, so the weights of the terms with small k, whose distance from the normal distribution is not necessarily small, tend to zero. Consequently, the whole expression tends to zero:

$$W_2^2(\gamma_n, Z) \longrightarrow 0 \qquad \text{as } n \to \infty.$$

This proves the theorem and gives a stronger convergence than the classical central limit theorem: distance has a quantitative meaning here and converges monotonically to zero.

To understand binary decomposition better, suppose that $n = 13$:

$$13 = 8 + 4 + 1,$$

$$\longrightarrow \quad \gamma_{13} = \frac{X_1 + \cdots + X_{13}}{\sqrt{13}}.$$

Consider the following binary blocks:

$$\eta_3 = \frac{X_1 + \cdots + X_8}{\sqrt{8}}, \qquad \eta_2 = \frac{X_9 + \cdots + X_{12}}{\sqrt{4}}, \qquad \eta_0 = X_{13}.$$

Then

$$\gamma_{13} = \sqrt{\frac{8}{13}}\eta_3 + \sqrt{\frac{4}{13}}\eta_1 + \sqrt{\frac{1}{13}}\eta_0.$$

$\longleftarrow$ Smaller distance $\longrightarrow$ Smaller weight